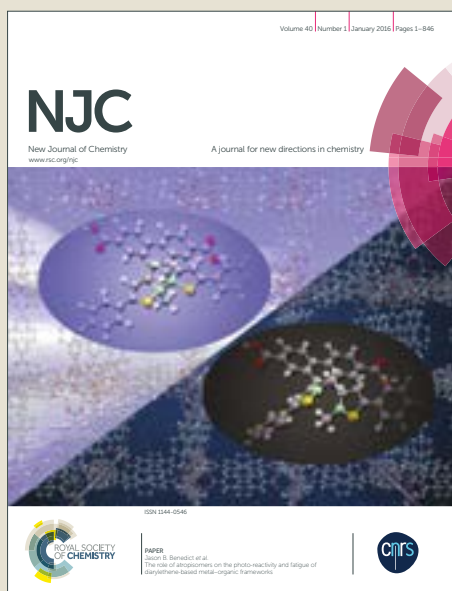


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Template-assisted in-situ confinement synthesis of nitrogen and oxygen co-doped 3D porous carbon network for high-performance sodium-ion batteries anode

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Non-graphitic carbons have shown great advantages as anode for sodium ion batteries, however, they deliver an unsatisfactory capacity especially at high rate due to the sluggish sodiation kinetics. In this work, we synthesized well-distributed nitrogen and oxygen co-doped three-dimensional ultrathin amorphous porous carbon network via a simple NaCl template-assisted in-situ confinement pyrolysis strategy. The porous carbon network with oxygen-containing groups provides abundant room (surface area of 282.78 m² g⁻¹) for the storage of Na⁺ and good wettability for the sufficient contact of active material and electrolyte, the affluent pores and the large interlayer space offer smooth passage for the insertion of Na⁺ and the transportation of electrons, high-content nitrogen (N: 12.44 at%) doping affords more defects and active sites for the redox capacitance reaction of Na⁺. When used as anode for sodium-ion batteries, the as-prepared sample presents high reversible capacity (416 mAh g⁻¹ at 0.1 A g⁻¹ after 100 cycles), superior rate capability (213.8 mAh g⁻¹ at 5A g⁻¹), and excellent cycling performance at super-high rate (142 mAh g⁻¹ at 10 A g⁻¹ after 1000 cycles with capacity retention of 94%). This work provides a new strategy to effectively construct continuous porous carbon nanostructure with uniform dual heteroatoms doping for high-performance sodium-ion batteries anode.

Introduction

With the continuous development of energy-storage power sources, sodium ion batteries (SIBs) came into people's view again after three decades, in that lithium ion batteries (LIBs) which is widely used in portable and smart devices is hard to satisfy the increasing demand of large-scale electric energy storage due to the limited lithium resource. The abundant reserves of Na (2.83% in Earth's crust and 30.6% in the ocean of Na vs only 20 ppm in Earth's crust of Li), popular price, only a slightly higher potential above Li (0.3 V of Na vs v of Li) and similar chemical characteristic compare with Li, making SIBs become a promising alternative to LIBs.¹⁻³ However, the electrode materials working in LIBs cannot display excellent electrochemical performance in SIBs due to the larger ionic diameter of Na⁺ (1.02 Å for Na⁺ vs. 0.76 Å for Li⁺) and different storage mechanism. Hence it's urgent to explore suitable electrode materials for SIBs with high capacity, good rate capability, and long cycle life.

To date, carbon-based materials (graphitic carbon, hard carbon, soft carbon), alloy materials (Sn, Sb, SnSb, and P), transitional metal oxides/sulfides (TiO₂, SnO₂, MoS₂), and organic materials have been employed as anode materials for SIBs.^{2,3} Among this, carbonaceous materials, especially disordered porous carbon materials with great interlayer distance have obtained particular concerned for its ascendant electrical conductivity, chemical stability and easy to bond with other non-carbon atoms. The porous nature of carbon not only provides sufficient contact area between electrode and electrolyte, but also offers adequate space to tolerate the volume change during Na⁺ de-intercalation process.

Furthermore, the introduction of heteroatoms (such as N, P, B, O, and S) into carbon anode of SIBs have been proved to be an effective approach to enhance the capacity, surface wettability, electronic conductivity of the materials, and then improve the charge transfer and redox reaction of electrode/electrolyte.⁴⁻⁷ Nitrogen is regarded as one of the most prospective dopants for carbon because of its higher electronegativity and similar atomic diameter compared to carbon. Studies of N-C system have indicated these chemical bonds (such as pyridine N (N-6), pyrrole N (N-5), graphite N (Graphitic N), quaternary nitrogen (N-Q) and so on) can change the surface structure of carbon material and add more ion implantation defects.⁸⁻¹⁰ Huan Liu¹¹ reported a nitrogen-rich carbon with interconnected porous structure via a nano-CaCO₃ template method, exhibiting a reversible capacity up to 338

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mAh g⁻¹ at a current density of 30 mA g⁻¹. Lei Wang¹² et al. synthesized porous N-doped carbon network composed with ultrathin carbon nanosheets via the direct pyrolysis of Vitamin B5, possessing a superior sodium storage: 335 mAh g⁻¹ at 100 mA g⁻¹ vs. Na/Na⁺. What's more, oxygen doping can improve the surface wettability of electrode and create more active sites for charge storage, which have been verified by increased pseudo capacitance of carbon electrode for supercapacitor.^{13,14} Thus it can be seen that the porous structure and N- as well as O- doping play important roles in improving the charge storage performance of carbon materials.

However, the nitrogen contents in these carbon materials are usually below 10.0 at% and the effect mechanism of N-/O- co-doping on Na-ion storage performance remains elusive. Besides, the common synthesis methods of N,O co-doped porous carbon such as chemical vapor deposition, thermal annealing with NH₃, nitrogen plasma treatment, arc-discharge, and the silica template methods, etc.¹⁵⁻¹⁹ suffer from tedious, complex, toxic, and the amount of nitrogen doped is hard to be adjusted. To address these problems, in-situ soluble template method have been put forward, taking NaCl, Na₂SiO₃, and Na₂CO₃ as template, have displayed great potential in fabricating porous carbon nanomaterials with heteroatoms doping due to the easily removal of template. Zhao^{20,21} and our group^{22,23} have successfully fabricated various 3D carbon networks and their composites by space-confinement of 3D NaCl template. However, a porous carbon network with high-content nitrogen/oxygen co-doping anode material with excellent Na-ion storage capability has never been fabricated by this method. Furthermore, the influencing mechanism of nitrogen/oxygen co-doping on the Na⁺ storage in porous carbon materials needs to be investigated.

In this work, we adopted NaCl template-assisted in-situ confinement pyrolysis method to synthesize an N, O- co-doped 3D ultrathin porous amorphous carbon network as anode materials for SIBs. Benefiting from the space confinement of NaCl template, the N, O co-doped amorphous carbon network exhibits large interlayer distance, great specific surface area as well as moderate porosity, which can provide more active sites, optimized electronic and ionic diffusion path, and lead to a fast diffusion kinetics of sodium ions. As a result, NING-N electrode delivers a high reversible capacity of 416 mAh g⁻¹ at 100 mA g⁻¹ after 100 cycles and excellent high rate performance of 213.8 mAh g⁻¹ at 5A g⁻¹ and 133 mAh g⁻¹ at 10 A g⁻¹ after 1000 cycles. This work demonstrated that the unique 3D porous carbon structure with nitrogen/oxygen co-doping have great promising to be an anode for high performance SIBs.

Results and discussion

The synthesis process of N/O- co-doped 3D ultrathin porous amorphous carbon network (NING-N) is schematically illustrated in Fig.1. First, carbon source (citric acid), soluble salt template (sodium chloride), and nitrogen source (urea) were uniformly dispersed in deionized water to form a transparent and homogeneous solution. After that the solution was frozen into solid and lyophilized in vacuum freeze drier.

During this process, the NaCl were self-assembled to be a 3D aggregate and covered with citric acid and urea evenly. The freeze-drying procedure can help keeping 3D cubic texture without collapsing during the removing of H₂O. Then the citric acid and urea were pyrolyzed and converted into carbon with nitrogen atoms doping during thermal treatment process under A₂ and H₂. After NaCl was removed by washing, the interconnected porous carbon network with nitrogen/oxygen co-doping is formed. The carbon network is composed of ultrathin nanosheets generated by the space confinement of NaCl template. And the nitrogen content can be tuned by controlling the ratio of N: C.

The morphology and microstructure of NING-N were investigated by scanning electron microscopy (SEM) and transmission electron microscopy (TEM). As shown in Fig.2a, there are many bubble-like carbon cells with a diameter of about 1 to 2 μm linking with each other, forming a whole connected carbon network. The spherical carbon cell is inheriting from the cubic NaCl template structure. Fig.2b shows a transparent ultrathin carbon cell with a large pore, which can also be observed in TEM images (Fig.2c). The interconnected ultrathin carbon network contribute to an improved electric conductivity, the rich porous structure is benefit for tolerating the volume change during Na⁺ intercalation/deintercalation and facilitating the fast transmission of electrons and ions. As shown in Fig.2d, an amorphous carbon layer can be observed, which is in favor of accommodating large Na⁺ ion insertion and keeping structure stability. The elements distribution in TEM (Fig.2e) shows that C, N, and O are dispersed uniformly. The doping of N produces more defects and reaction activity sites for redox reaction of Na⁺; moreover, the presences of O is favorable for accessibility of electrolyte ions, and an excellent electrochemical performance is expected. The SEM and TEM images (Fig.S1) of carbon network prepared using glucose as carbon source (PU/PU-N) shows the similar morphology, which demonstrate that the formation of N-/O- co-doped carbon network is not effected by the carbon sources.

Fig.3a displays the XRD patterns of NING-N and NING. The diffraction patterns of NING-N and NING displayed only two broad peaks located at 25° and 44°, which are corresponding to (002) and (100) planes of graphite.^{24,25} The broaden peaks confirm a typical amorphous carbon with disordered structure, which can be ascribed to the low pyrolysis temperature. No other peak was observed in the spectrogram indicating no other substances or impurities in the samples.

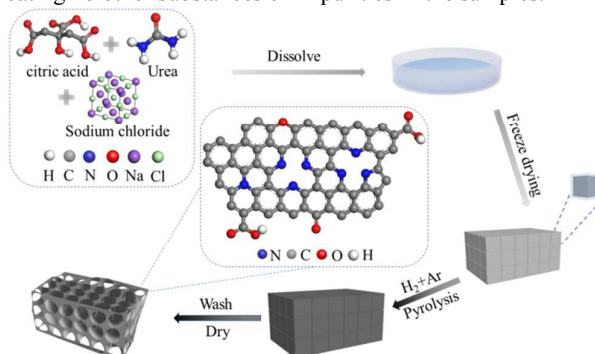


Fig.1 Schematic illustration of the preparation of N-/O- co-doping 3D porous carbon network (NING-N)

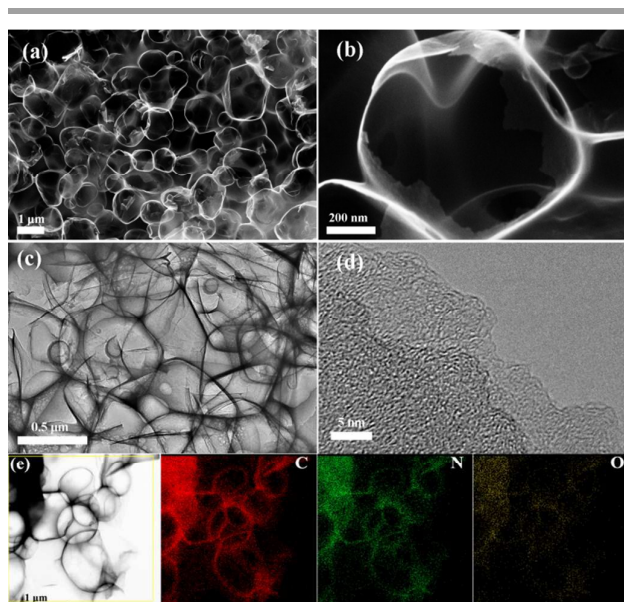


Fig.2 (a, b) SEM (c, d) TEM and (e) Elements mapping of NING-N.

A slight shift of C (002) diffraction is observed compared with graphite, indicating larger interlayer spacing in NING-N and NING. The XRD pattern of PU-N and PU are shown in Fig. S2a, the results are very similar with NING-N. The amorphous nature of NING-N and NING are examined by Raman spectra as shown in Fig.3b. The remarkable peaks at 1356.6 cm^{-1} and 1601 cm^{-1} are assigned to D mode and G mode.^{17,26} Commonly, D peak reflects the disorder structure of the material, while G peak is associated with sp^2 carbon and reflects the graphitization degree of materials. The intensity ratio I_D/I_G is 0.935 and 0.81 in NING-N and NING, the larger I_D/I_G value indicates higher degree of disorder of the material, which testified that N-doping creates more defects and increases the randomness of materials, which is in favor of Na^+ storage. The Raman spectra of PU-N and PU are shown in Fig.S2b, it is demonstrated that the PU-N and PU are also amorphous phase with high-content defects.

The specific surface area and pore structure of NING-N and NING (Fig.3c,d) are measured by nitrogen adsorption-desorption isotherms. The specific surface area of NING and NING-N calculated by Brunauer-Emmett-Teller (BET) equation are $282.78\text{ m}^2\text{ g}^{-1}$ and $265.61\text{ m}^2\text{ g}^{-1}$, respectively. The nitrogen adsorptions desorption curves of NING-N/NING belong to a typical type-IV isotherm, and the pronounced hysteresis loop reveal mesopores structure. The pore size calculated by Barrett-Joyner-Halenda method (BJH) is in the range of 1 to 10 nm, indicating the coexisting of micropores and mesopores. The total pore volume is calculated to be $0.286\text{ cm}^3\text{ g}^{-1}$, and the pore volume values of micropores and mesopores are $0.125\text{ cm}^3\text{ g}^{-1}$ and $0.161\text{ cm}^3\text{ g}^{-1}$, respectively. The rich porous structures are also obtained in PU-N and PU

(Fig.S2c, d), which is conducive to the accessibility of electrolyte.

The elemental composition and chemical bonding of NING-N were determined by XPS measurement. As shown in Fig.4a, the sample NING-N is composed of C, O, and N atoms, indicating that the heteroatoms N and O have been successfully incorporated into carbon frameworks. The atomic ratio of C, N and O in NING-N are calculated to be 80.88 %, 12.44%, and 6.68%, respectively. The XPS spectra of PU-N are shown in Fig.S3, which present a similar shape compare with NING-N, and the corresponding data for PU-N are 90.34 %, 4.75 %, and 4.91% (Table S1). This demonstrates that the both oxygen and nitrogen functional groups were successfully preserved in carbon network through the in-situ pyrolysis method. It should be noted that there are some differences in the relative content of nitrogen and oxygen in NING-N and PU-N, which indicates that the heteroatom doping content can be moderately regulated with the proportion of doping sources and pyrolysis conditions. Fig.4b shows the magnified fitting of $\text{C}1\text{s}$ spectrum for NING-N, the peaks located at 284.71, 285.9, 286 and 289 eV are related to C-C, C-N, C-O, and C=O bonds, respectively. The magnified $\text{O}1\text{s}$ spectra can be fitted into three peaks in Fig.4d, which are recognized as oxygen in the COOH/CO (531.1eV), C=O (532 eV), and C-O (533.eV), suggesting the formation of some carboxyl and lactone groups.²⁷ These oxygen functional groups are benefit for the electrolyte wettability of porous electrode, leading to enhancement of diffusion kinetics. The N 1s peaks (Fig.4c) at 398.4, 400.21, 400.86 eV correspond to pyridinic nitrogen (N-6), pyrrolic nitrogen (N-5), and graphite nitrogen, respectively.²⁸⁻³⁰ As shown in Fig.4c, the proportion of pyridine nitrogen is the highest. The research results show that pyridinic nitrogenous doping is helpful to enhance the adsorption capacity of materials, which can improve the capacity of sodium ion storage of materials. The graphite nitrogen contributes to forming a rich electron structure in NING-N, which can improve the conductivity of carbon. Such high-content atomic doping ratios of O- and N- groups is benefit for providing more defects and improving the surface Na-ion storage (capacitive performance) of 3D porous carbon networks. To sum up, the large surface area, abundant porous structure together with high content N-doping are benefit for facilitating the fast diffusion of Na^+ and transportation of electrons, O-doping increases the wettability of electrode and electrolyte and then improves the Na^+ storage capacity effectively.

The electrochemical performances of the electrodes were evaluated with 2032-type coin cells. Cyclic voltammetry curves were used to study the redox reaction of NING-N and NING (Fig.5a, c) in the range of 0.005-3V at scanning rate of 0.1 mV s^{-1} . There're significant differences in CV curves between the first cycle and the second cycle, which is attributed to the formation of a solid electrolyte interphase (SEI) layer and Na^+ irreversible embedding. After the first cycle, the subsequent CV curves are almost overlapped, indicating high reversibility of Na-ion intercalation or de-intercalation. Both the CV curves of NING-N and NING exhibit sharp peaks at low potential (0.005-0.2 V) and some humps between 0.2 V and 1.5 V, the sharp peaks at low potential could be attributed to Na^+ insertion and de-intercalation in the interlayer of carbon, and

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the humps at high potential can be ascribed to Na^+ adsorption/desorption or charge storage on the surface of the electrode (pseudocapacitance).

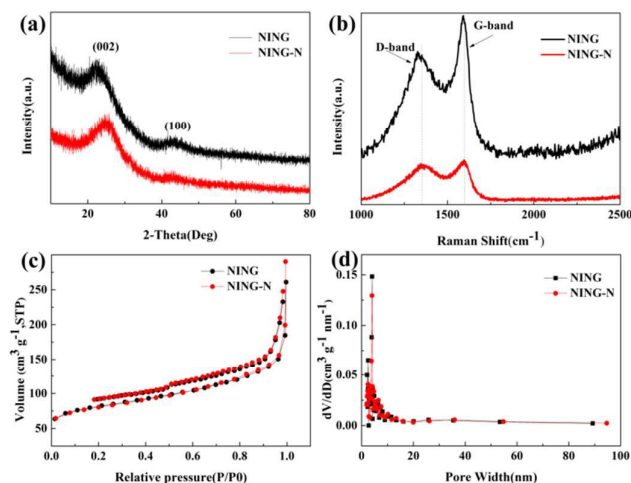


Fig. 3 (a) XRD; (b) Raman spectra; (c) Adsorption and desorption isothermal curves; (d) Desorption pore volume distribution of NING-N and NING.

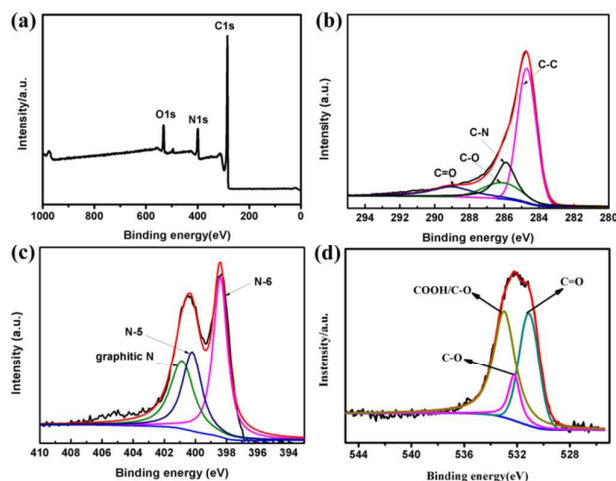


Fig. 4 (a) XPS survey spectra, (b-d) high resolution C1s, N1s, and O1s spectra of NING-N.

Fig.5b and Fig.5d show the voltage profiles of NING-N and NING, an apparent long slope and a small plateau below 0.2V in both discharge curves are observed, demonstrating the continuous insertion/de-insertion of Na^+ and surface charge storage/adsorption desorption mechanism of Na^+ , which is consistent with the CV curves. During the subsequent cycles the curves are almost overlapped, revealing an excellent cyclic stability of materials. The first charge-discharge specific capacity is 402.1 mAh g⁻¹ and 1241.4 mAh g⁻¹ for NING-N, corresponding to a low coulombic efficiency (~37 %), which is caused by the irreversible side reaction associated with large surface area. However, once a stable solid electrolyte interphase (SEI) is formed, the coulombic efficiency increases approaching to 100 % in the subsequent cycles. After 100 cycles

under 0.1 A g⁻¹, NING-N can still delivers a high reversible capacity of 416 mAh g⁻¹ without capacity fading but a slight increases, as shown in Fig.5e. The increased capacity with cycling can be ascribe to the activation process of the porous carbon anode. A longer discharge plateau and higher capacity of NING-N compare with that of NING attest that N doping is in favor of Na^+ storage and improve the reversible capacity of the materials. Even so, a stable cycle performance of NING with high coulombic efficiency cannot be ignore, which proving the porous carbon network is in favor of maintaining structure stable stability during repeated cycles.

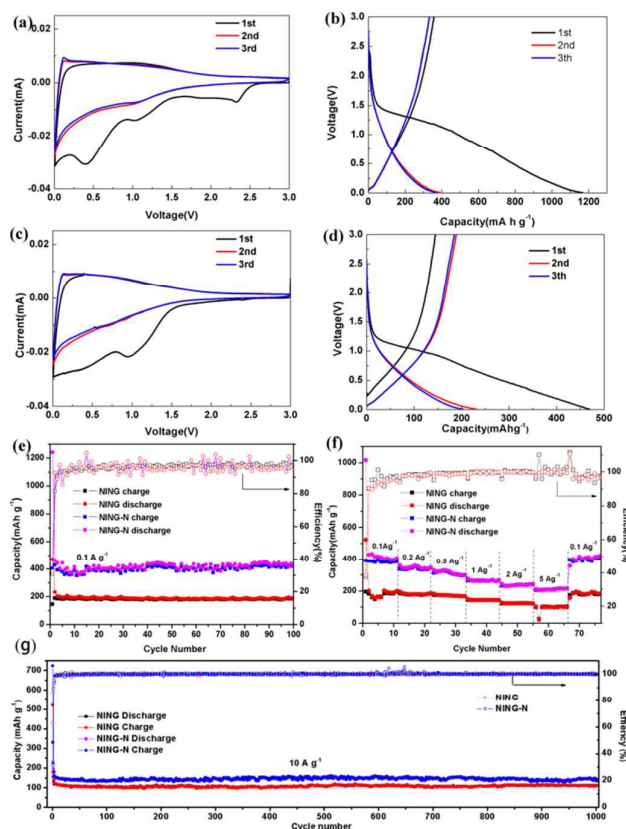


Fig.5 The CV curves of NING-N and NING, (b, d) Charge/discharge curves of NING-N and NING for the initial 3 cycles. (e) Cycling performance of NING-N and NING at 0.1 A g⁻¹ for 100 cycles. (f) Rate capability of NING-N and NING at various rates; (g) Cycling performance of NING-N and NING at 10 A g⁻¹ for 1000 cycles.

Fig.5f shows the rate capability of NING-N and NING, in which the current density was increased from 0.1 to 5 A g⁻¹ and then returned to 0.1 A g⁻¹. The specific capacity of NING-N is 402, 333, 296, 264, 240, and 213.8 mAh g⁻¹ under the current density of 0.1, 0.2, 0.5, 1, 2 and 5 A g⁻¹. There's only a slightly capacity fading along with the increasing of current density. The splendid rate performance is due to the superiority of N/O- co-doped porous carbon network structure. Besides, when the current density is reset to 0.1 A g⁻¹, a high specific capacity of 409 mAh g⁻¹ is restored, which shows a perfect electrochemical reversibility. And it is notable that the specific capacities of NING-N are always higher than that of

NING at any current density, which proved an enhanced rate performance are achieved by N-/O- co-doping. As can be seen from Fig.5g, the capacity retention for NING-N is still as high as 94 % even after 1000 cycles at 10 A g⁻¹, indicating that NING-N has a superior electrochemical cycling stability at super-high rate as anode material for SIBs.

The excellent stability for NING-N is owing to the presence of unique pore structure, which allowed the penetrating of electrolyte and provided plenty of favorable sites for Na ions diffusion. Moreover, the existence of heteroatoms may improve the electrochemical performance by the enhancement of wettability and diffusion kinetics, which can be supported by the EIS measurements (Fig. S5). The electrochemical performance of PU and PU-N used as anode for SIBs are shown in Fig.S4. The N-/O- co-doped sample PU-N also showed improved Na⁺ storage performance than PU. In addition, the as-prepared carbon materials without adding NaCl template exhibit a block-like structure in micrometer scale rather than a 3D porous structure, resulting in a poor performance for Na⁺ storage (Fig.S6). Table S2 showed the comparison for Na⁺ storage performance with other heteroatom-doped carbon in the literatures. It is obvious that the N-/O- co-doped porous carbon in this work deliver much higher capacity especially at large current density and superior cycling performance compared with other heteroatom-doped carbon materials.

In order to further investigate to the electrochemical behavior of NING-N, the CV curves at different scan rates (from 0.2 to 5 mV s⁻¹) were carried out as shown in Fig.6a. Obviously, there is a pair of faradaic redox peaks between 0 and 1.5 V, indicating the presence of redox reactions associated with Na⁺ intercalation and surface atoms redox reaction. The shapes of CV curves are similar at various scan rates, suggesting the apparent pseudocapacitance behavior and excellent rate capability. Moreover, with the increase of the scan rates, the anodic peaks move forward higher potential while the cathodic peaks shift to lower potential due to polarization. To sum up, the charge-discharge mechanisms can be considered to diffusion-controlled faradaic contribution from Na⁺ intercalation reaction, the charge storage at surface atoms (pseudocapacitance effect), and the non-faradaic contribution from electrical double layer effect.³¹⁻³³ In the CV curves (Fig.6a), the current is divided based the equation $i = k_1 v + k_2 v^{1/2}$, where the k_1 represents capacitive process and the k_2 means diffusion-controlled process.^{34,35} The redox pseudocapacitance-like contribution was qualitatively distinguished according to the equation and the CV curves. Fig.6b shows that the $i/v^{1/2}$ is linearly related to the $v^{1/2}$, and the slope is k_1 . As a result, the contribution of capacitive process for NING-N is 80% at 5 mVs⁻¹ (Fig.6c), which is consistent with the better rate capability of NING-N. Fig.6d summarizes the percentage of the capacitive contribution for NING-N at different scan rates. Such high capacitive contributions for NING-N at high scan rates are related to their high surface area, good electrical conductivity, and high-content nitrogen doping for abundant active sites, which account for the excellent rate capability.

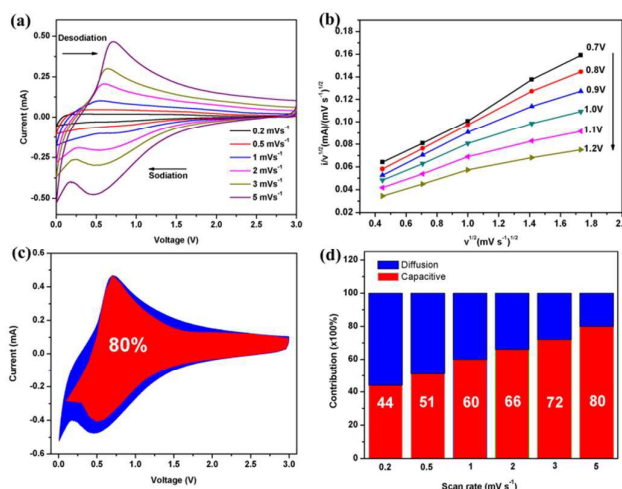


Fig. 6 (a) CV curves at various scan rates (from 0.2 mV s⁻¹ to 5 mV s⁻¹); (b) $i/v^{1/2}$ versus $v^{1/2}$; (c) Capacitive (red) and diffusion-controlled (blue) contribution to the charge storage at 5 mV s⁻¹; (d) Contribution ratio of capacitive (red) and diffusion-controlled (blue) capacities at different scan rates.

Experimental

Synthetic procedures

Citric acid (1.25 g), urea (1.25 g) and NaCl (20.64 g) were dissolved in deionized water to form a homogeneous mixture solution, and then the solution was frozen in a refrigerator for 48 h. Put the sample in a freeze dryer for 24 h to sublimate the water. The powder sample was collected and annealed at 750 °C in an atmosphere of argon and hydrogen for 2h. Wash the sample with deionized water for several times to remove the NaCl template, after drying in a vacuum oven at 100 °C, an N,O co-doping porous carbon was obtained, named as NING-N. In order to inspect the effect of N-doped on the electrochemical performance, a sample without N-doping synthesized in the same method was prepared, named as NING. To investigate the effect of carbon sources, porous carbon with glucose as the carbon source was also prepared, the samples were named as (PU and PU-N), which can be found in the supporting information.

Materials and methods

Scanning electron microscopy (FESEM, ZEISS, SUPRA-55), transmission electron microscopy (TEM, JEM 2100F) were used to investigate the morphology of samples. X-ray diffraction (XRD, Rigaku MiniFlexII DX-2500, Cu K α radiation) and X-ray photoelectron spectroscopy (XPS, Thermo escalab 250XI) were used to analysis the structure and composition of the samples. Raman spectroscopy (Renishaw, InVia microscope, a 532 nm laser) was used to determine the defects and crystallinity of the samples. Nitrogen absorption desorption isothermal was used to analysis the specific surface area and pore distribution.

Electrochemical measurements

The electrochemical performances of the materials were tested using 2032-type coin cells with Na metal as the counter and reference electrode. As-prepared 3D ultrathin amorphous porous carbon

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network (NING-N, NING, PU, PU-N), carbon black and PVDF (polyvinylidene fluoride) were stirred in a weight ratio of 80:10:10 (wt. %) in NMP (N-Methyl pyrrolidone) to form slurry. Coat the slurry on copper foil substrates and dried at 60°C for 8 hours. The mass loading of the electrodes is about 1.0 mg cm⁻². Finally, the N-ion batteries were assembled in a glove box in which the concentration of moisture and oxygen were maintained at <0.1 ppm. The electrolyte was 1 M NaClO₄ dissolved in 1:1 EC (ethylene carbonate)/DEC (diethyl carbonate) and 5% wt. FEC (fluoroethylene carbonate). Cyclic voltammetry (CV) were carried out using electrochemical workstation (CHI 660E) at different scan rate from 0.1 mV s⁻¹ to 5 mV s⁻¹ in a voltage window from 0.01 to 3.0 V. The galvanostatic charge-discharge tests were carried out on a LAND2001A battery testing system at varied current rates over a voltage range of 0.005 to 3.00 V.

Conclusions

In conclusion, we have developed a novel in-situ NaCl template-assisted pyrolysis and space confinement strategy for the fabrication of a well-distributed N, O co-doped 3D ultrathin porous amorphous carbon network, using citric acid or glucose as carbon source, and urea as nitrogen sources. Profiting from the unique 3D porous structure with large surface area, interconnected amorphous carbon network with large interlayer distance, and high-content N, O doping induced more active sites and superior diffusion kinetics of sodium ion, NING-N shows high reversible capacity (416 mAh g⁻¹ at 0.1 A g⁻¹ after 100 cycles), good rate capability (213.8 mAh g⁻¹ at 5 A g⁻¹), and superior cycling performance at super-high rate (142 mAh g⁻¹ at 10 A g⁻¹ after 1000 cycles with capacity retention of 94%). Furthermore, the unique carbon architecture with uniform N, O dual heteroatoms doping can be potentially extended to be utilized in super capacitors, catalysis, air electrode, and so forth.

Author contributions

D. Wang conceived the idea, analyzed the data, and wrote the main manuscript text. Y. Li and K. Z. Dong characterized the electrochemical performance of the samples. S. H. Luo, Y. G. Liu, and X. W. Qi provided advice for the research. Z. Y. Wang supervised and conceived this study, and provided intellectual and technical guidance. All authors discussed the results and commented on the manuscript.

Conflicts of interest

There are no conflicts to declare.

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References

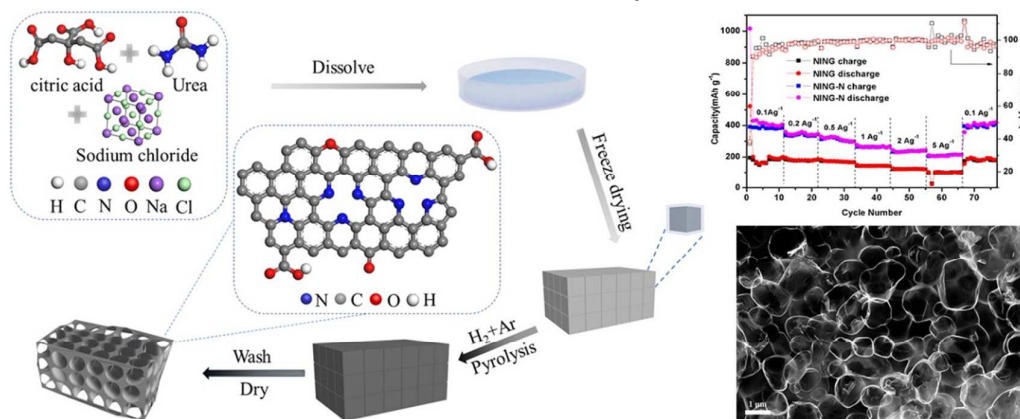
- 1 M. D. Slater, D. Kim, E. Lee and C. S. Johnson, *Adv. Funct. Mater.*, 2013, 23, 947-958.
- 2 W. Luo, F. Shen, C. Bommier, H. Zhu, X. Ji and L. Hu, *Acc. Chem. Res.*, 2016, 49, 231-240.
- 3 J. Y. Hwang, S. T. Myung and Y. K. Sun, *Chem. Soc. Rev.*, 2017, 46, 3529-3614.
- 4 K. P. S. S. Hembram, H. Jung, B. C. Yeo, S. J. Pai, S. Kim, K. R. Lee and S. S. Han, *J. Phys. Chem. C*, 2015, 119, 15041-15046.
- 5 Y. Li, Z. Wang, L. Li, S. Peng, L. Zhang, M. Srinivasan and S. Ramakrishna, *Carbon*, 2016, 99, 556-563.
- 6 J. P. Paraknowitsch and A. Thomas, *Energy & Environ. Sci.*, 2013, 6, 2839-2855.
- 7 L. Zhang, G. Xia, Z. Guo, X. Li, D. Sun and X. Yu, *Int. J. Hydrogen Energy*, 2016, 41, 14252-14260.
- 8 Z. Zhou, X. Gao, J. Yan, D. Song and M. Morinaga, *Carbon*, 2004, 42, 2677-2682.
- 9 J. C. Yong, S. K. Han, H. Im, Y. Myung, G. B. Jung, W. L. Chi, J. Park, M. H. Park, J. Cho and S. K. Hong, *J. Phys. Chem. C*, 2011, 115, 9451-9457.
- 10 C. Ma, X. Shao and D. Cao, *J. Mater. Chem.*, 2012, 22, 8911-8915.
- 11 H. Liu, M. Jia, N. Sun, B. Cao, R. Chen, Q. Zhu, F. Wu, N. Qiao and B. Xu, *Acs Appl. Mater. Interfaces*, 2015, 7, 27124-27130.
- 12 L. Wang, C. Yang, S. Dou, S. Wang, J. Zhang, X. Gao, J. Ma and Y. Yu, *Electrochim. Acta*, 2016, 219, 592-603.
- 13 W. Feng, P. He, S. Ding, G. Zhang, M. He, F. Dong, J. Wen, L. Du and M. Liu, *Rsc Adv.*, 2016, 6, 5949-5956.
- 14 L. Zhang, L. Xu, Y. Zhang, X. Zhou, L. Zhang, A. Yasin, L. Wang and K. Zhi, *Rsc Adv.*, 2018, 8, 3869-3877.
- 15 M. Wang, Y. Yang, Z. Yang, L. Gu, Q. Chen and Y. Yu, *Adv. Sci.*, 2017, 4, 1600468-1600475.
- 16 H. Hou, X. Qiu, W. Wei, Y. Zhang and X. Ji, *Adv. Eng. Mater.*, 2017, 7, 1602898-1602927.
- 17 D. Qin, Z. Liu, Y. Zhao, G. Xu, F. Zhang and X. Zhang, *Carbon*, 2018, 130, 664-671.
- 18 Z. Wang, Q. Long, L. Yuan, W. Zhang, X. Hu and Y. Huang, *Carbon*, 2013, 55, 328-334.
- 19 D. Li, H. Chen, G. Liu, M. Wei, L. X. Ding, S. Wang and H. Wang, *Carbon*, 2015, 94, 888-894.
- 20 J. Qin, C. He, N. Zhao, Z. Wang, C. Shi, E. Z. Liu and J. Li, *Acs Nano*, 2014, 8, 1728-1738.
- 21 S. Zhu, J. Li, Q. Li, C. He, E. Liu, F. He, C. Shi and N. Zhao, *Electrochim. Acta*, 2016, 212, 621-629.
- 22 Z. Wang, D. Wang, S. Luo, S. Bao, Y. Liu, X. Qi, C. He, C. Shi and N. Zhao, *New J. Chem.*, 2016, 41.
- 23 Z. Wang, S. Luo, F. Chen, D. Wang, Y. Liu, X. Qi, C. Shi and N. Zhao, *Rsc Adv.*, 2016, 6, 54718-54726.
- 24 Y. Cao, L. Xiao, M. L. Sushko, W. Wang, B. Schwenzer, J. Xiao, Z. Nie, L. V. Saraf, Z. Yang and J. Liu, *Nano Letters*, 2012, 12, 3783-3787.
- 25 K. Tang, L. Fu, R. J. White, L. Yu, M. M. Titirici, M. Antonietti and J. Maier, *Adv. Energy Mater.*, 2012, 2, 873-877.
- 26 Y. Mao, H. Duan, B. Xu, L. Zhang, Y. S. Hu, C. Zhao, Z. Wang, L. Chen and Y. Yang, *Energy Environ. Sci.*, 2012, 5, 7950-7955.

Journal Name

ARTICLE

- 27 Z. N. Tetana, S. D. Mhlanga and N. J. Coville, *Diamond Relat. Mater.*, 2017, 74, 70-80.
- 28 G. Zhou, E. Paek, G. S. Hwang and A. Manthiram, *Nat. Commun.*, 2015, 6, 7760-7770.
- 29 Y. Liu, N. Zhang, L. Jiao and J. Chen, *Adv. Mater.*, 2015, 27, 6702-6707.
- 30 Z. Tian, M. Xiang, J. Zhou, L. Hu and J. Cai, *Electrochim. Acta*, 2016, 211, 225-233.
- 31 B. K. Lesel, J. S. Ko, B. Dunn and S. H. Tolbert, *Acs Nano*, 2016, 10, 7572-7581.
- 32 Z. Chen, V. Augustyn, X. Jia, Q. Xiao, B. Dunn and Y. Lu, *Acs Nano*, 2012, 6, 4319-4327.
- 33 G. A. Muller, J. Cook, H. S. Kim, S. H. Tolbert and B. Dunn, *Nano Letters*, 2015, 15, 1911-1917.
- 34 V. Augustyn, P. Simon and B. Dunn, *Energy Environ. Sci.*, 2014, 7, 1597-1614.
- 35 D. Chao, C. Zhu, P. Yang, X. Xia, J. Liu, J. Wang, X. Fan, S. V. Savilov, J. Lin and H. J. Fan, *Nat. Commun.*, 2016, 7, 12122-12129.

Table of contents entry



Nitrogen and oxygen co-doped porous carbon network is synthesized by NaCl template-assisted in-situ confinement method and used for high-performance sodium-ion batteries anode.